

CHAPTER 2

Energy

Energy makes things happen. It runs your phone, your body, and every star in the sky.

 Easy Read Edition

WHAT WE WILL LEARN

- ★ What is energy?
- ★ Types of energy
- ★ Energy of motion (KE)
- ★ Stored energy (PE)
- ★ How energy switches
- ★ Energy is never lost
- ★ Heat and temperature
- ★ Three ways heat moves

2.1 What Is Energy?

You eat lunch. You feel full of energy. Then you can run, jump, and think.

Energy is what makes things happen. Without energy, nothing moves. Nothing changes. Nothing works.

We measure energy in **joules** (J). One joule is small. One Calorie of food has 4,184 joules!



Battery
has energy



Sun
has energy



Food
has energy

2.2 Types of Energy

Energy comes in many forms. Here are the main ones.

Type of Energy	What It Is	Example
Kinetic	Energy of motion	A bike rolling
Potential	Stored energy	A book on a shelf
Thermal (Heat)	Energy in hot things	A cup of cocoa
Chemical	Energy in food and fuel	A battery, food
Electrical	Energy in moving charges	Lightning, plug
Light	Energy you can see	Sunlight, screens
Sound	Energy in vibrations	Music, voices
Nuclear	Energy in atoms	The Sun's core

★ Cool Fact

Your lunch is full of energy from the Sun. Plants stored sunlight as chemical energy. Then you ate the plants (or animals who ate them). Your food is bottled sunshine!

2.3 Kinetic Energy (Motion)

Kinetic energy is the energy of motion. Anything moving has kinetic energy. We call it **KE** for short.

Two things change how much KE something has:

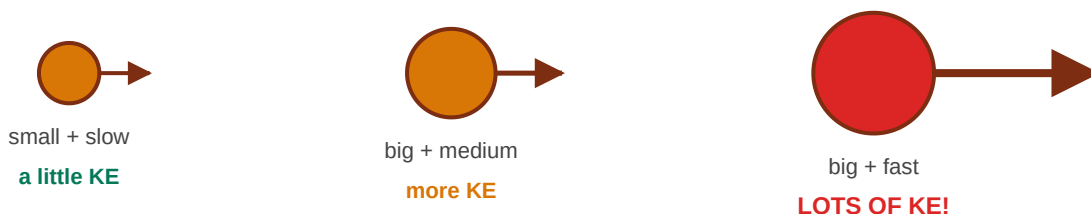
1. Mass

More mass = more KE. A bowling ball has more KE than a tennis ball at the same speed.

2. Speed

More speed = MUCH more KE. Speed matters a LOT. Doubling speed makes 4 times the KE!

More mass OR more speed = more KE



Bigger and faster = more kinetic energy.

2.4 Potential Energy (Stored)

Potential energy is stored energy. It is waiting to be used. We call it **PE** for short.

Think of a roller coaster at the top of a hill. It is not moving. But it has lots of stored energy. About to be released!

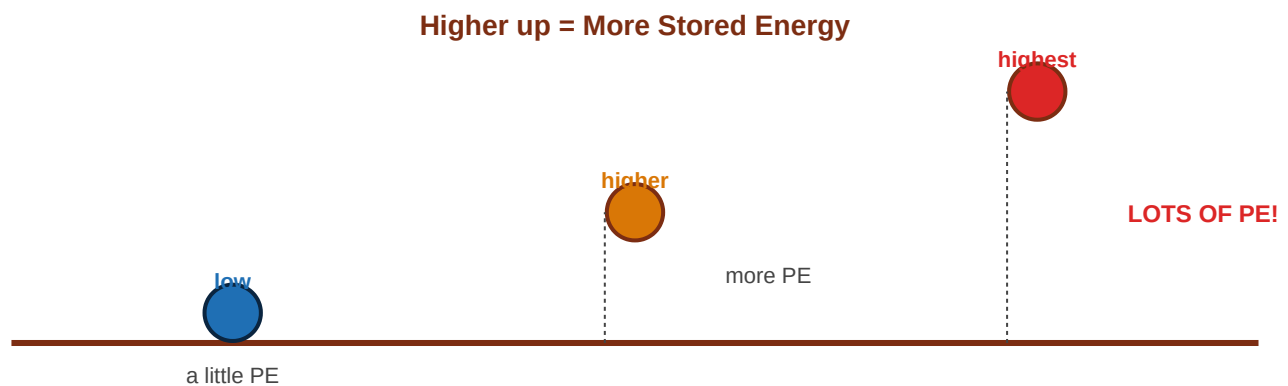
Two things change PE:

1. Mass

More mass = more PE. A heavy object up high has lots of stored energy.

2. Height

More height = more PE. The higher up something is, the more PE it has.



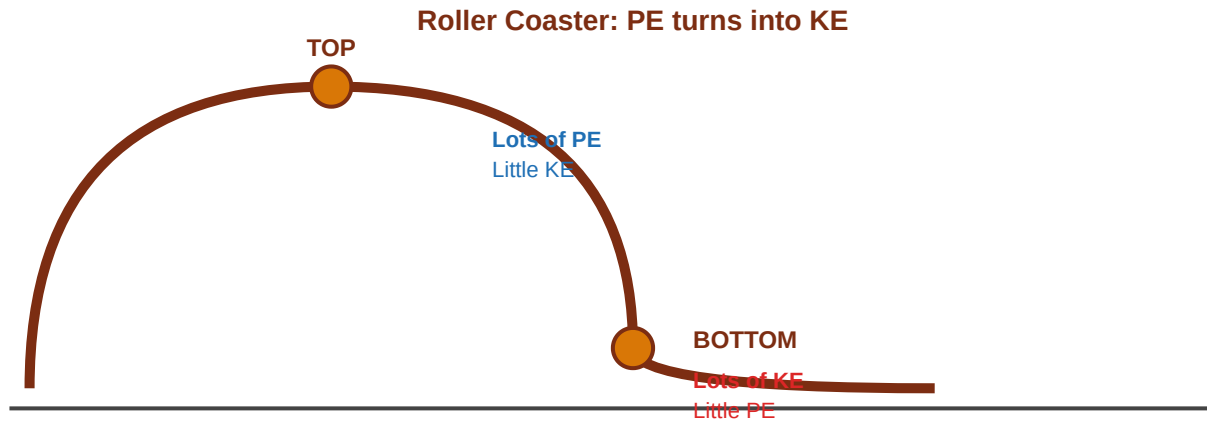
The higher up an object is, the more PE it has.

2.5 KE and PE Switch Back and Forth

Here is something cool: KE and PE trade places.

Think about a roller coaster:

- **Top of hill:** Lots of PE. Not moving much. Little KE.
- **Going down:** PE turns into KE. Speeds up!
- **Bottom:** Lots of KE. Moving fast! Little PE.



PE at the top turns into KE at the bottom.

Other things that switch KE and PE:

- A swing going back and forth
- A stretched rubber band snapping back
- A bouncing ball

2.6 Energy Changes Forms

Energy can change from one form to another. This is called an **energy transformation**.

These happen all around you, all the time!

Object	Energy Changes From → To
A match	Chemical → Light + Heat
A light bulb	Electrical → Light + Heat
A speaker	Electrical → Sound
Solar panel	Light → Electrical
A flashlight	Chemical → Electrical → Light
A car engine	Chemical → Heat → Motion
You eating food	Chemical → Motion + Heat

2.7 Energy Is Never Lost



The Big Rule

Energy cannot be made or destroyed. It can only change form.

This is called the **Conservation of Energy**.

So why does a bouncing ball stop bouncing?

It didn't lose energy. The energy turned into heat and sound from rubbing on the floor and the air. The energy is still there — it just spread out.



Cool Example

Rub your hands together fast. Feel them get warm? You just turned motion energy into heat energy. Energy doesn't disappear — it just changes form.

2.8 Heat and Temperature

Heat and temperature are NOT the same thing!

Temperature

How fast the particles are moving on average. Measured in °F or °C.

Heat (Thermal Energy)

The TOTAL energy of all the moving particles. Depends on temperature AND the amount of stuff.



Watch Out!

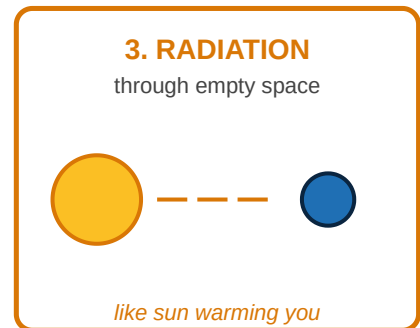
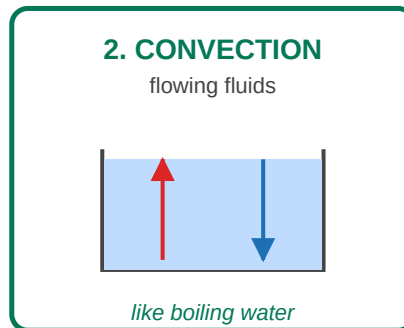
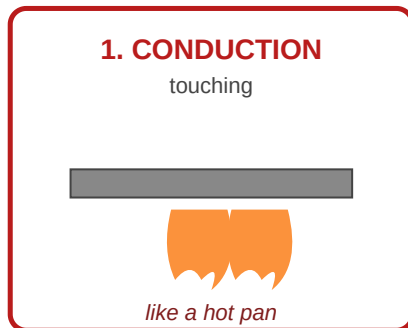
A cup of boiling water has high temperature. But a big bathtub of warm water has more total heat. Why? Because there is more water — more particles to add up!

Heat always moves from **hot** to **cold**. Always. Never the other way!

When you grab an ice cube, you don't "get cold." Your heat flows INTO the ice. The ice melts because YOUR heat went into it.

2.9 Three Ways Heat Moves

Heat moves in three different ways.



Three ways heat can move from one place to another.

1. Conduction (touching)

Heat moves through things that touch. A hot pan handle warms your hand. Works best in solids like metal.

2. Convection (flowing)

Hot stuff rises. Cold stuff sinks. This makes currents in liquids and gases. Like boiling water or warm air rising.

3. Radiation (through space)

Heat travels as waves. It does NOT need any matter to move through! That's how the Sun's heat reaches Earth through empty space.

★ All Three at Once

A campfire shows all three ways at once!

- Touch a hot stick from the fire = **conduction**
- Hot air rising from flames = **convection**
- Warmth on your face = **radiation**

Check Yourself

1. Which way does heat travel from the Sun?
2. Which way is a hot pan handle warming your hand?
3. Which way is boiling water mixing itself up?

★ CHAPTER 2 — BIG IDEAS

- **Energy** makes things happen. It's measured in **joules**.
- **Kinetic energy** is motion. It depends on mass and speed.
- **Potential energy** is stored. It depends on mass and height.
- **KE and PE trade places** in things like roller coasters and swings.
- Energy changes from one form to another all the time.
- **Energy is never made or destroyed**. It only changes form.
- **Temperature** is how fast particles move. **Heat** is total energy.
- Heat moves three ways: **conduction, convection, and radiation**.